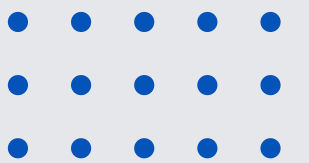


Anna Garcia

Determining the Cause of Death

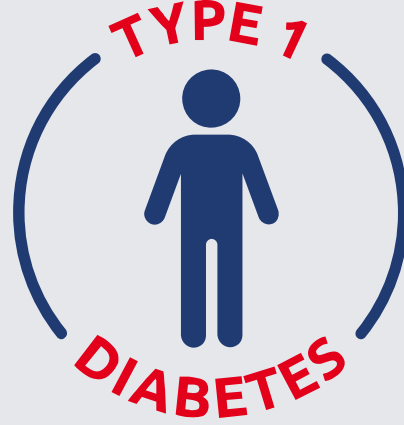


Forensic Science

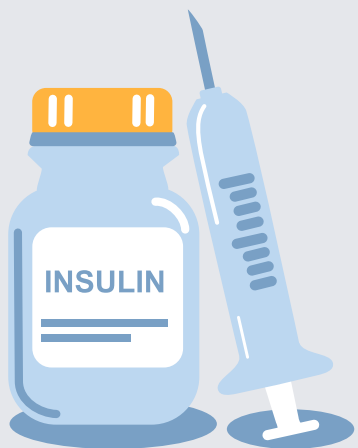
- Anna Garcia was found dead in her home. External abrasions on her head suggested a fall, but no foul play was indicated.
- DNA from the scene matched Anna. There were no unknown DNA profiles present.
- Initial investigations ruled out trauma as a primary cause of death.
- Final cause of death determined to be natural causes, not homicide.



Diabetes

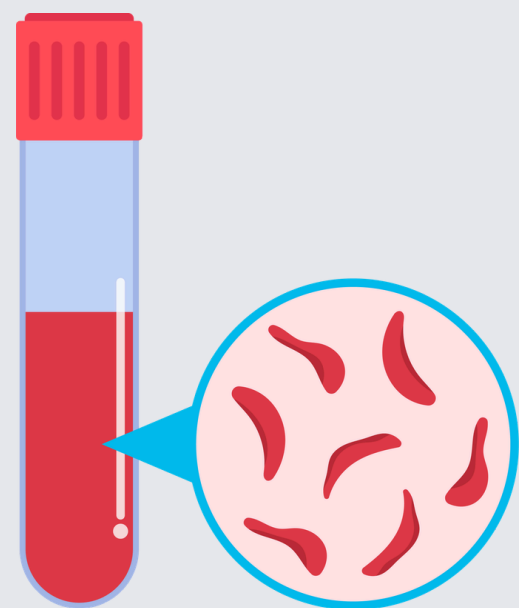
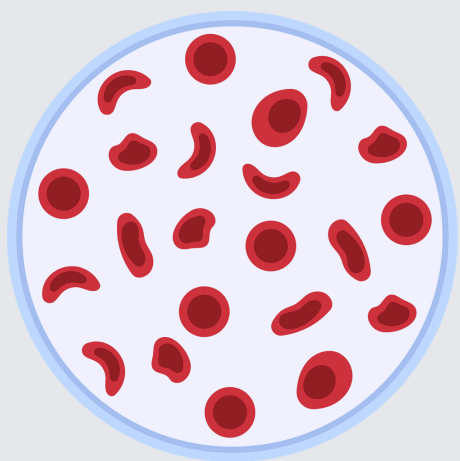


- Symptoms & Medical History:
 - Frequent urination, excessive thirst, weight loss, and fatigue.
 - Glucose detected in urine.
 - Glucose tolerance test and HbA1c confirmed Type 1 diabetes.
 - Tingling in feet may signal early diabetic neuropathy.
- Diagnosis Reasoning:
 - Body can't process glucose → burns fat/muscle instead → weight loss.
 - High blood sugar → kidney filters excess → frequent urination and thirst.



Sickle Cell Anemia

- Symptoms & Medical History:
 - Diagnosed in infancy.
 - Sickle cell crisis → extreme pain in legs → hospitalized.
 - Chronic anemia and fatigue present.
 - Required catheter → later led to UTI.
- Diagnosis Reasoning:
 - Misshapen red blood cells → block vessels → pain, poor circulation.
 - Anemia caused by destruction of abnormal cells.



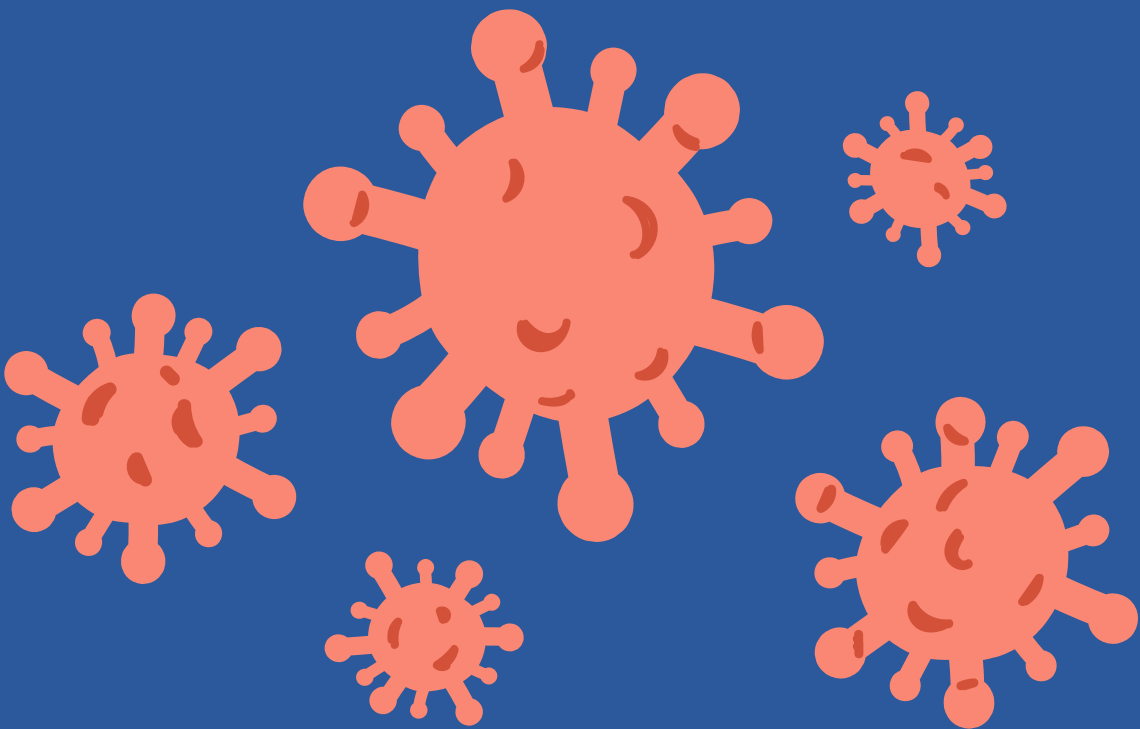
Heart Disease

- Symptoms & Medical History:
 - Two chest pain incidents.
 - Diet high in fat, no formal exercise.
 - Total cholesterol: 389 mg/dL (very high)
 - LDL: 243 mg/dL → risk for atherosclerosis.
 - Family history of heart disease and hypertension.
- Diagnosis Reasoning:
 - High LDL → plaque buildup → narrowed arteries.
 - Chest pain likely due to reduced oxygen to heart (angina).
 - Risk for heart attack or stroke due to occluded vessels.



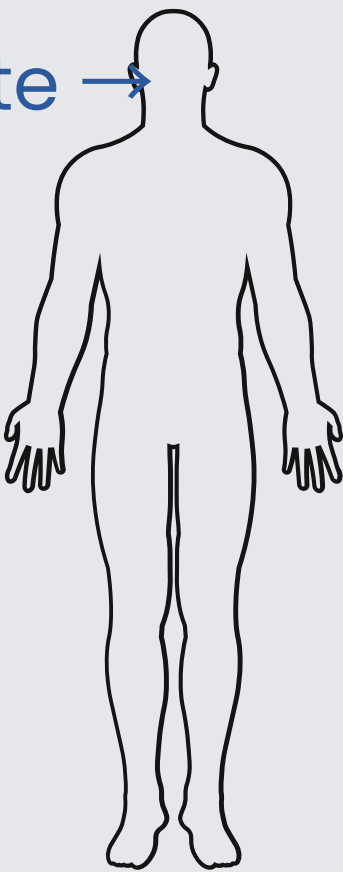
Infectious Disease

- Symptoms & Medical History:
 - Post-hospitalization: persistent urge to urinate, cloudy/pink urine → signs of UTI.
 - Positive urinalysis for bacterial infection.
 - Likely due to catheter use during sickle cell crisis.
- Diagnosis Reasoning:
 - Hospital-acquired UTI from catheter.
 - Infection could cause spleen enlargement (splenomegaly) or inflammation.



Body as a whole

- Circulatory: Heart pumps blood → vessels transport nutrients/oxygen.
- Respiratory: Lungs exchange gases → oxygen to blood, CO₂ out.
- Digestive: Breaks down food → absorbs nutrients → waste removed.
- Nervous: Sends electrical signals for movement, reflexes, thoughts.
- Endocrine: Glands release hormones → control metabolism, growth.
- Urinary: Filters blood → removes waste → regulates water/salt.



Cause of Death Breakdown

- Sick Cell: Occluded splenic artery → infarction, necrosis.
- Infection: Likely enlarged/inflamed spleen.
- Diabetes: Weakened blood vessels, impaired healing.
- Heart Disease: Weakened heart, poor response to blood loss.



Comparison to final autopsy

- How my causes fit the official autopsy:
 - Splenic artery occlusion aligns with sickle cell and diabetes complications.
 - Enlarged spleen = consistent with sickle cell and/or infection.
 - Internal bleeding + anemia = low RBC, hemoglobin, and hematocrit.
 - Heart strain noted → likely struggled to compensate during hemorrhage.
- Evidence
 - CBC (low RBC, HCT, HGB)
 - 500 mL internal bleeding
 - Abdominal swelling
 - Splenomegaly.





Final Determination: Natural Causes



Unit	Cause of Death	Manner	Reasoning
Diabetes	Artery damage → splenic occlusion	Natural	Diabetes causes vessel damage, poor healing.
Heart Disease	Heart strain during bleed	Natural	High cholesterol and cardiac stress worsened bleeding response.
Sickle Cell	Blocked artery, spleen necrosis	Natural	Sickle cells block flow → infarction.
Infection	Splenomegaly from UTI → rupture risk	Natural	Infection enlarged spleen → more fragile.